

L03 Taylor's Theorem

CS 169 / CS 268
Introduction to Optimization

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Approximations and their errors

Fix a reference point x_0 . Write $x = x_0 + h$, so $h = x - x_0$ is the displacement.

First order: keep the constant and linear terms.

Second order: also keep the quadratic terms.

Signed error = actual value – approximation.

R_1 and R_2 denote the first- and second-order errors. Their absolute values give the error sizes.

Taylor's theorem: first order (1D)

Univariate. Suppose f is differentiable at an interior point x_0 .

$$f(x_0 + h) = f(x_0) + f'(x_0)h + R_1(h).$$

Equivalently, set $h = x - x_0$:

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + R_1(x - x_0).$$

$$\lim_{h \rightarrow 0} \frac{|R_1(h)|}{|h|} = 0, \quad R_1(h) = o(|h|).$$

First-order error: a bound

Assume f has two continuous derivatives on an open interval containing x_0 and $x_0 + h$.

Error formula. For some point ξ between the endpoints,

$$R_1(h) = \frac{1}{2}f''(\xi)h^2.$$

If $|f''(x)| \leq M_2$ throughout the segment, then

$$|R_1(h)| \leq \frac{M_2}{2}|h|^2.$$

First order: exponential example

Let $f(x) = e^x$ and $x_0 = 0$. Then $f(0) = f'(0) = 1$.

$$e^h \approx 1 + h, \quad e^{0.1} \approx 1.1.$$

$$R_1(0.1) = e^{0.1} - 1.1 \approx 0.005170918.$$

$$|R_1(0.1)| \leq \frac{e^{0.1}}{2} (0.1)^2 < 0.00553.$$

Taylor's theorem: second order (1D)

Univariate. Suppose f is C^2 near x_0 : its first two derivatives are continuous.

$$f(x_0 + h) = f(x_0) + f'(x_0)h + \frac{1}{2}f''(x_0)h^2 + R_2(h).$$

Equivalently, set $h = x - x_0$:

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{1}{2}f''(x_0)(x - x_0)^2 + R_2(x - x_0).$$

$$\lim_{h \rightarrow 0} \frac{|R_2(h)|}{|h|^2} = 0, \quad R_2(h) = o(|h|^2).$$

Second-order error: a bound

Assume f has three continuous derivatives on an open interval containing x_0 and $x_0 + h$.

Error formula. For some point ξ between the endpoints,

$$R_2(h) = \frac{f'''(\xi)}{6} h^3.$$

If $|f'''(x)| \leq M_3$ throughout the segment, then

$$|R_2(h)| \leq \frac{M_3}{6} |h|^3.$$

Second order: exponential example

For $f(x) = e^x$ at $x_0 = 0$, we also have $f''(0) = 1$.

$$e^h \approx 1 + h + \frac{h^2}{2}, \quad e^{0.1} \approx 1.105.$$

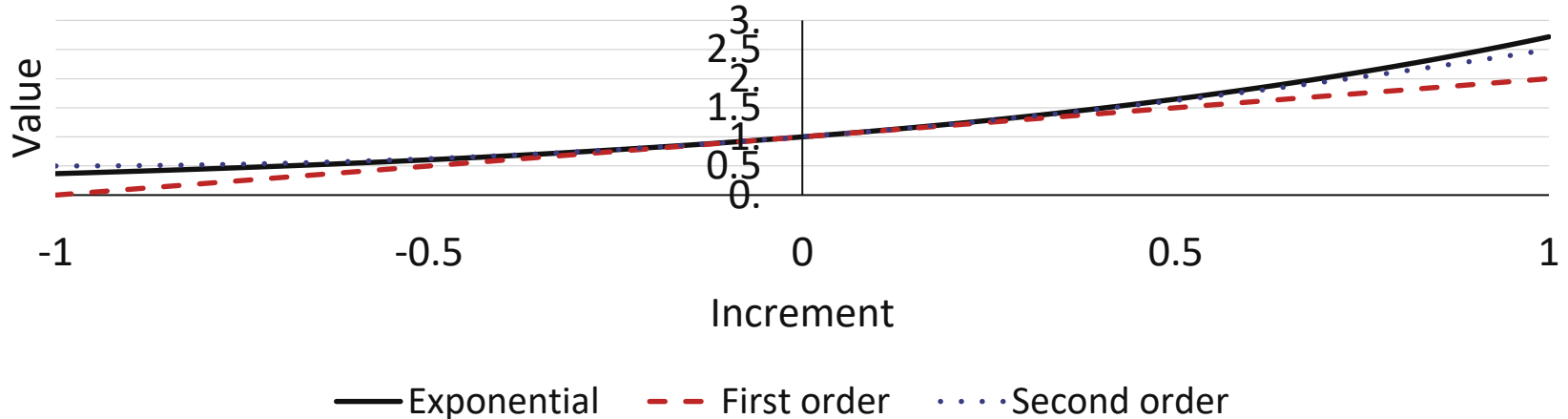
$$R_2(0.1) = e^{0.1} - 1.105 \approx 0.000170918.$$

$$|R_2(0.1)| \leq \frac{e^{0.1}}{6} (0.1)^3 < 0.000185.$$

Comparing the exponential approximations

First order: $1 + x$, Second order: $1 + x + \frac{x^2}{2}$.

For $f(x) = e^x$ near 0, adding the quadratic term improves the approximation.



A Taylor approximation need not give a global bound.

Smaller steps, smaller errors

$$R_1(h) = e^h - 1 - h, \quad R_2(h) = e^h - 1 - h - \frac{h^2}{2}.$$

$$R_1(0.1) \approx 0.0051709, \quad R_1(0.05) \approx 0.0012711.$$

$$R_2(0.1) \approx 0.000170918, \quad R_2(0.05) \approx 0.0000210964.$$

Here, halving h reduces the first-order error by about 4 and the second-order error by about 8.

Gradient descent: one variable

Suppose f is differentiable at an interior point x_0 and $f'(x_0) \neq 0$. Choose a step size $\eta > 0$.

$$h = -\eta f'(x_0), \quad x_1 = x_0 + h.$$

$$f(x_1) - f(x_0) = -\eta [f'(x_0)]^2 + R_1(h).$$

Here $R_1(h) = o(\eta)$ as $\eta \rightarrow 0^+$: the error divided by η tends to zero.

For sufficiently small $\eta > 0$, the negative term dominates, so $f(x_1) < f(x_0)$.

Vectors, gradients, and step size

Now $x_0, x, h \in \mathbb{R}^n$, with $x = x_0 + h$. The displacement is $h = x - x_0$.

$$\|h\|_2 = \sqrt{h_1^2 + \cdots + h_n^2}.$$

$$\nabla f(x_0)^\top h = \sum_{i=1}^n \frac{\partial f}{\partial x_i}(x_0) h_i.$$

A small step means $\|h\|_2$ is small, from any direction.

Taylor's theorem: first order (nD)

Multivariate. Suppose f is differentiable at x_0 in an open domain of $\mathbb{R}^n$.

$$f(x_0 + h) = f(x_0) + \nabla f(x_0)^\top h + R_1(h).$$

Equivalently, set $h = x - x_0$:

$$f(x) = f(x_0) + \nabla f(x_0)^\top (x - x_0) + R_1(x - x_0).$$

$$\lim_{h \rightarrow 0} \frac{|R_1(h)|}{\|h\|_2} = 0, \quad R_1(h) = o(\|h\|_2).$$

First order: a two-variable example

$$f(x_1, x_2) = (x_1 - 1)^2 + 2(x_2 + 1)^2.$$

$$x_0 = (2, 0), \quad f(x_0) = 3, \quad \nabla f(x_0) = \begin{pmatrix} 2 \\ 4 \end{pmatrix}.$$

$$h = (0.1, -0.1) : \quad f(x_0 + h) \approx 3 + 0.2 - 0.4 = 2.8.$$

$$f(x_0 + h) = 2.83, \quad R_1(h) = 2.83 - 2.8 = 0.03.$$

The Hessian matrix

Definition. For a C^2 function, the Hessian contains the second partial derivatives.

$$H_f(x_0) = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2}(x_0) & \frac{\partial^2 f}{\partial x_1 \partial x_2}(x_0) \\ \frac{\partial^2 f}{\partial x_2 \partial x_1}(x_0) & \frac{\partial^2 f}{\partial x_2^2}(x_0) \end{pmatrix}.$$

C^2 means that the second partial derivatives exist and are continuous near the reference point.

Then the mixed partials agree, so the Hessian is symmetric.

Reading the quadratic term

For a symmetric 2×2 matrix H ,

$$h^\top H h = H_{11}h_1^2 + 2H_{12}h_1h_2 + H_{22}h_2^2.$$

Quadratic term. Taylor's formula uses one half of this expression:

$$\frac{1}{2}h^\top H h = \frac{H_{11}}{2}h_1^2 + H_{12}h_1h_2 + \frac{H_{22}}{2}h_2^2.$$

Taylor's theorem: second order (nD)

Multivariate. Suppose f is C^2 on an open neighborhood of $x_0 \in \mathbb{R}^n$.

$$f(x_0 + h) = f(x_0) + \nabla f(x_0)^\top h + \frac{1}{2}h^\top H_f(x_0)h + R_2(h).$$

Equivalently, set $h = x - x_0$:

$$f(x) = f(x_0) + \nabla f(x_0)^\top (x - x_0) + \frac{1}{2}(x - x_0)^\top H_f(x_0)(x - x_0) + R_2(x - x_0).$$

$$\lim_{h \rightarrow 0} \frac{|R_2(h)|}{\|h\|_2^2} = 0, \quad R_2(h) = o(\|h\|_2^2).$$

Second order: the two-variable example

For the same quadratic and $x_0 = (2, 0)$,

$$H_f(x_0) = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}.$$

$$f(x_0 + h) = 3 + 2h_1 + 4h_2 + h_1^2 + 2h_2^2.$$

$$h = (0.1, -0.1) : \quad f(x_0 + h) = 2.83, \quad R_2(h) = 0.$$

A coupled three-variable example

$$f(x_1, x_2, x_3) = x_1^2 + x_2^2 + x_3^2 + x_1 x_2 x_3.$$

$$x_0 = (1, 1, 1), \quad f(x_0) = 4.$$

$$\nabla f(x_0) = \begin{pmatrix} 3 \\ 3 \\ 3 \end{pmatrix}.$$

The product term couples all three coordinates.

The Hessian in three variables

Differentiate each entry of the gradient once more.

$$H_f(x) = \begin{pmatrix} 2 & x_3 & x_2 \\ x_3 & 2 & x_1 \\ x_2 & x_1 & 2 \end{pmatrix} \quad H_f(x_0) = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}.$$

The Hessian is symmetric, but it changes with the point x .

Taylor approximations in three variables

First-order approximation.

$$f(x_0 + h) \approx 4 + 3(h_1 + h_2 + h_3).$$

Second-order approximation.

$$\begin{aligned} f(x_0 + h) \approx & 4 + 3(h_1 + h_2 + h_3) \\ & + h_1^2 + h_2^2 + h_3^2 \\ & + h_1h_2 + h_1h_3 + h_2h_3. \end{aligned}$$

The exact errors in three variables

Exact value = second-order approximation + $h_1h_2h_3$.

First-order error.

$$R_1(h) = h_1^2 + h_2^2 + h_3^2 \\ + h_1h_2 + h_1h_3 + h_2h_3 + h_1h_2h_3.$$

Second-order error.

$$R_2(h) = h_1h_2h_3.$$

A numerical three-variable step

$$h = (0.1, 0.1, 0.1), \quad x_0 + h = (1.1, 1.1, 1.1).$$

$$f(x_0 + h) = 3(1.1)^2 + (1.1)^3 = 4.961.$$

First order: 4.9, Second order: 4.96.

$$R_1(h) = 0.061, \quad R_2(h) = 0.001.$$

Halving the three-variable step

$$h = (0.05, 0.05, 0.05), \quad f(x_0 + h) = 4.465125.$$

First order: 4.45, Second order: 4.465.

$$R_1(h) = 0.015125, \quad R_2(h) = 0.000125.$$

First-order error: reduced by about 4.

Second-order error: reduced by exactly 8.

Gradient descent: several variables

Suppose f is differentiable at an interior point x_0 and $\nabla f(x_0) \neq 0$. Choose a step size $\eta > 0$.

$$h = -\eta \nabla f(x_0), \quad x_1 = x_0 + h.$$

$$f(x_1) - f(x_0) = -\eta \|\nabla f(x_0)\|_2^2 + R_1(h).$$

Here $R_1(h) = o(\eta)$ as $\eta \rightarrow 0^+$: the error divided by η tends to zero.

For sufficiently small $\eta > 0$, the negative term dominates, so $f(x_1) < f(x_0)$.

Taylor's theorem: summary

First order. Keep the value and linear derivative terms. Differentiability gives $R_1(h) = o(\|h\|_2)$.

Second order. The Hessian adds the quadratic correction; C^2 gives $R_2(h) = o(\|h\|_2^2)$.

Error. Always subtract the approximation from the actual value. Exact formulas or derivative bounds give more information.

Next: linear optimization

A linear objective has no quadratic correction. Linear constraints determine the feasible choices.

$$\begin{aligned} &\text{maximize } 3x_1 + 2x_2 \\ &\text{subject to } x_1 + x_2 \leq 4, \\ &\qquad\qquad 0 \leq x_1 \leq 2, \quad x_2 \geq 0. \end{aligned}$$

Next week: modeling linear programs and the simplex method.

Reading: Section 5.6. **Review:** Section 3.4.

Questions to check understanding

1. Does Taylor's theorem guarantee that the second-order approximation is more accurate at every point? Why does the distance from x_0 matter?
2. Why does a sufficiently small step opposite a nonzero gradient decrease the function? What can go wrong with a large step?